$$BD^2 = a^2 + d^2 - 2ad\cos\theta$$
$$BD^2 = b^2 + c^2 - 2bc\cos(180^\circ - \theta)$$

$$a^2 + d^2 - 2ad\cos\theta = b^2 + c^2 + 2bc\cos\theta$$

$$\boxed{\cos\theta = \frac{a^2 + d^2 - b^2 - c^2}{2(ad + bc)}}$$

$$Q_1 = \frac{1}{2}ad\sin\theta$$
$$Q_2 = \frac{1}{2}bc\sin(180^\circ - \theta)$$

$$Q = Q_1 + Q_2 = \frac{1}{2}(ad + bc)\sin\theta$$

$$\boxed{\sin\theta = \frac{2Q}{ad + bc}}$$

$$\because \sin^2\theta + \cos^2\theta = 1$$

$$\therefore \left(\frac{2Q}{ad + bc}\right)^2 + \left(\frac{a^2 + d^2 - b^2 - c^2}{2(ad + bc)}\right)^2 = 1$$
$$\implies \boxed{16Q^2 = 4(ad + bc)^2 - (a^2 + d^2 - b^2 - c^2)^2}$$

---

$$\begin{aligned}
16Q^2 &= \left(2(ad + bc) + a^2 + d^2 - b^2 - c^2\right)\left(2(ad + bc) - a^2 - d^2 + b^2 + c^2\right) \\
&= \left((a + d)^2 - (b - c)^2\right)\left((b + c)^2 - (a - d)^2\right) \\
&= (a + d - b + c)(a + d + b - c)(b + c - a + d)(b + c + a - d) \\
&= 16(s - a)(s - b)(s - c)(s - d)
\end{aligned}$$

$$\boxed{Q^2 = (s - a)(s - b)(s - c)(s - d)}$$

---

For a triangle with sides $a, b, c$, let $d = 0$

$$Q^2 = s(s - a)(s - b)(s - c)$$

$$Q = \sqrt{s(s - a)(s - b)(s - c)}$$

---

This solution was submitted by **Jon**

Published and refined by **Jon**

*Last updated: July 2, 2026*